Machine Learning (CAI-262)

Objectives:

- **Course Learning Objectives (CLOs)**
- **What is Machine Learning**
- **Difference between AI and ML**
- **Types of Machine Learning**
- **Main Challenges of Machine Learning**

Course Learning Objectives (CLOs)

CLO-1: Describe basic machine learning concepts, theories, and applications.

CLO-2: Apply supervised learning techniques to solve classification problems of moderate complexity.

CLO-3: Apply unsupervised learning techniques to solve clustering problems of moderate complexity.

Reference material:

1. Machine Learning, Tom, M., McGraw Hill, 1997.
2. Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, MIT Press,2012.
3. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. by Aurélien Géron. Released September 2019. Publisher(s): O'Reilly Media, Inc.

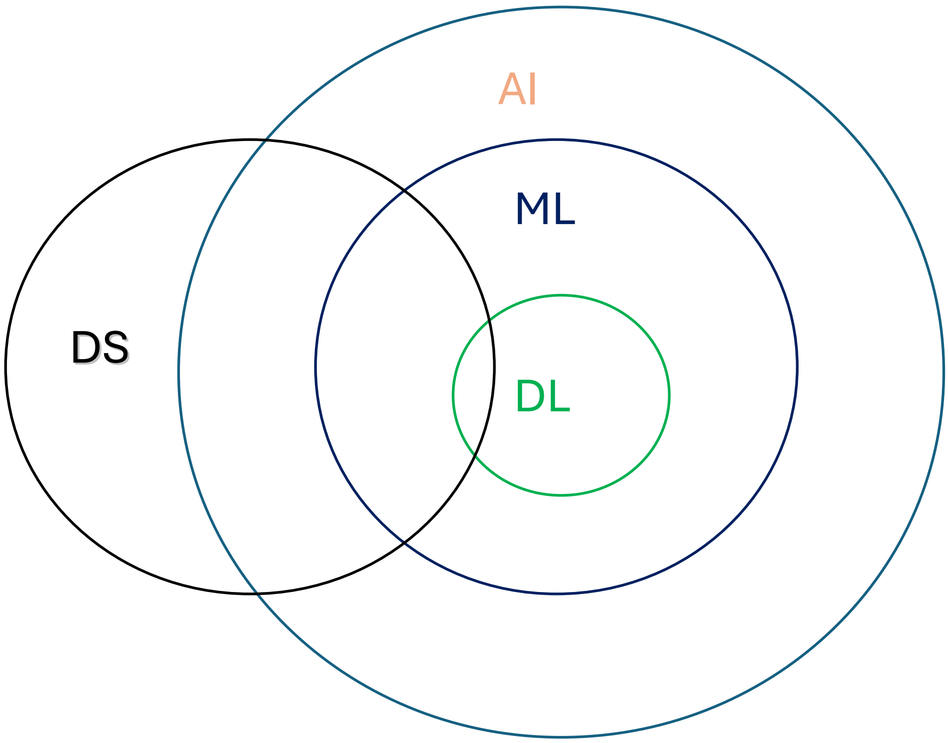
Roadmap for AI domain Subjects

Sr.#	Course Code	Prerequisite	Subject	Credit hours
1	CSC-203		Artificial Intelligence	3 (2-3)
2	CAI-261		Programming for AI	3 (2-3)
3	CAI-262	CSC-203	Machine Learning	3 (2-3)
4	CAI-361	CAI-261	Artificial Neral Networks	3 (2-3)
5	CAI-362		Knowledge Representation and Reasoning	3(2-3)
6	CAI-363	CAI-361	Computer Vision	3 (2-3)
7	CAI-364	CAI-361	Natural Language Processing	3 (2-3)
8	CAI-461	CAI-262	Reinforcement Learning	3 (2-3)

Machine Learning

Machine Learning is the science (and art) of programming computers so they can learn from data. Here is a slightly more general definition:

> **“Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.” (Arthur Samuel, 1959)**



Artificial Intelligence (AI)

Machine Learning (ML)

Deep Learning (DL)

Data Science (DS)

Difference between AI and ML

Making a machine (computer) perform the same tasks which you have just done is called **Artificial Intelligence** and if you learn to do these tasks using existing data, then this is called **Machine Learning.**

So Machine Learning is a technique to learn patterns from the data and make predictions. The machine learning algorithms implementation is data-based.

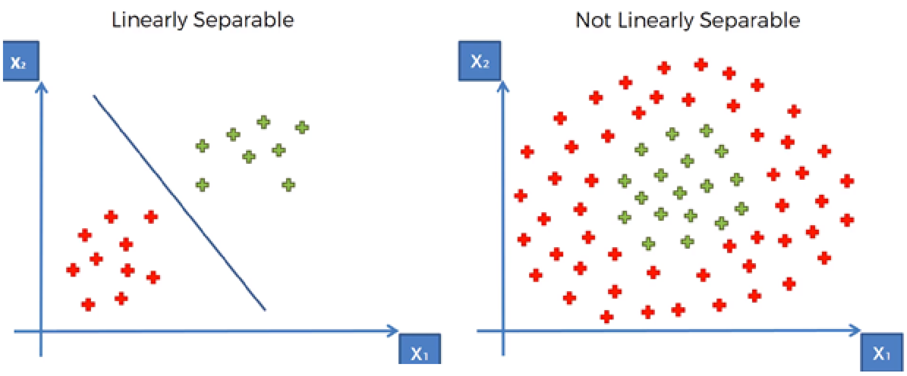
Types of Machine Learning

Supervised Learning

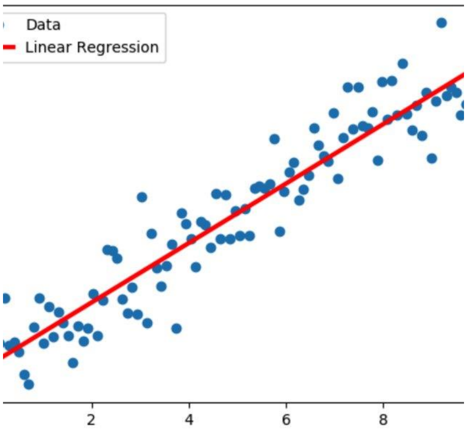
Supervised learning uses a training set to teach models to yield the desired output. This training dataset includes inputs and correct outputs, which allow the model to learn over time. The algorithm measures its accuracy through the loss function, adjusting until the error has been sufficiently minimized.

Supervised learning can be separated into two types of problems: classification and regression.

Classification uses an algorithm to accurately assign test data into specific categories. It recognizes specific entities within the dataset and attempts to draw some conclusions on how those entities should be labelled or defined. Common classification algorithms are linear classifiers, support vector machines (SVM), decision trees, k-nearest neighbor, and random forest.

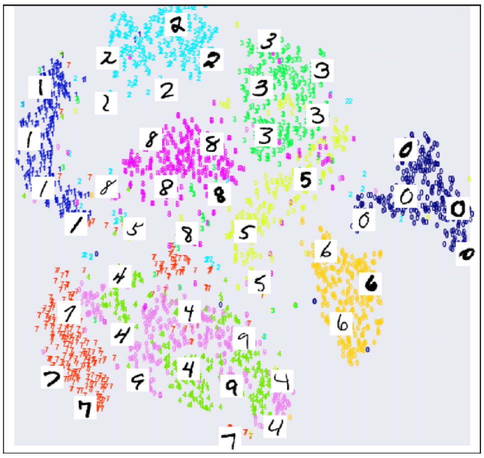


Regression is used to understand the relationship between dependent and independent variables. It is commonly used to make projections, such as for sales revenue for a given business. Linear regression, logistical regression, and polynomial regression are popular regression algorithms.



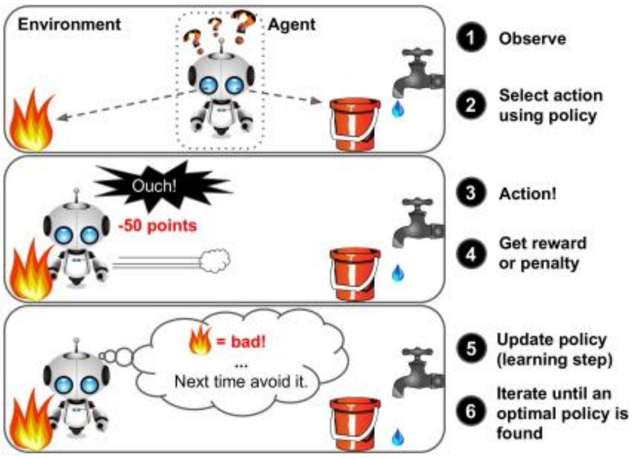
Unsupervised Learning

Unlike supervised learning, unsupervised learning uses unlabelled data. From that data, it discovers patterns that help solve for **clustering** or association problems. This is particularly useful when subject matter experts are unsure of common properties within a data set. Common clustering algorithms are hierarchical, k-means, and Gaussian mixture models.



Reinforcement Learning

Reinforcement Learning is a very different approach. The learning system, called an agent in this context, can observe the environment, select and perform actions, and get rewards in return (or penalties in the form of negative rewards). It must then learn by itself what is the best strategy, called a policy, to get the most reward over time. A policy defines what action the agent should choose when it is in a given situation.



Differnces among Supervised, Unsupervised and Reinforcement Learning

Supervised Learning	Unsupervised Learning	Reinforcement Learning
Data provided is labeled data, with output values specified	Data provided is unlabeled data, the outputs are not specified, machine makes its own prediction	The machine learns from its environment using rewards and errors
Used to solve Regression and classification problems	Used to solve Association and clustering problems	Used to solve Reward based problems
Labeled data is used	Unlabeled data is used	No predefined data is used
External Supervision	No supervision	No supervision
Solves problems by mapping labeled input to known output	Solves problems by understanding patterns and discovering output	Follows Trail and Error problem solving approach

Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Nonrepresentative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data